A Project Review on

Enhancing Skin Disease Classification with MetaCLIP Models: A Multi-Dataset Analysis Across 15 Sources

Under the guidance of

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Flow of Presentation

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- Dataset description
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- Proposed Algorithm
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  - Classification report
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  - Accuracy Comparison
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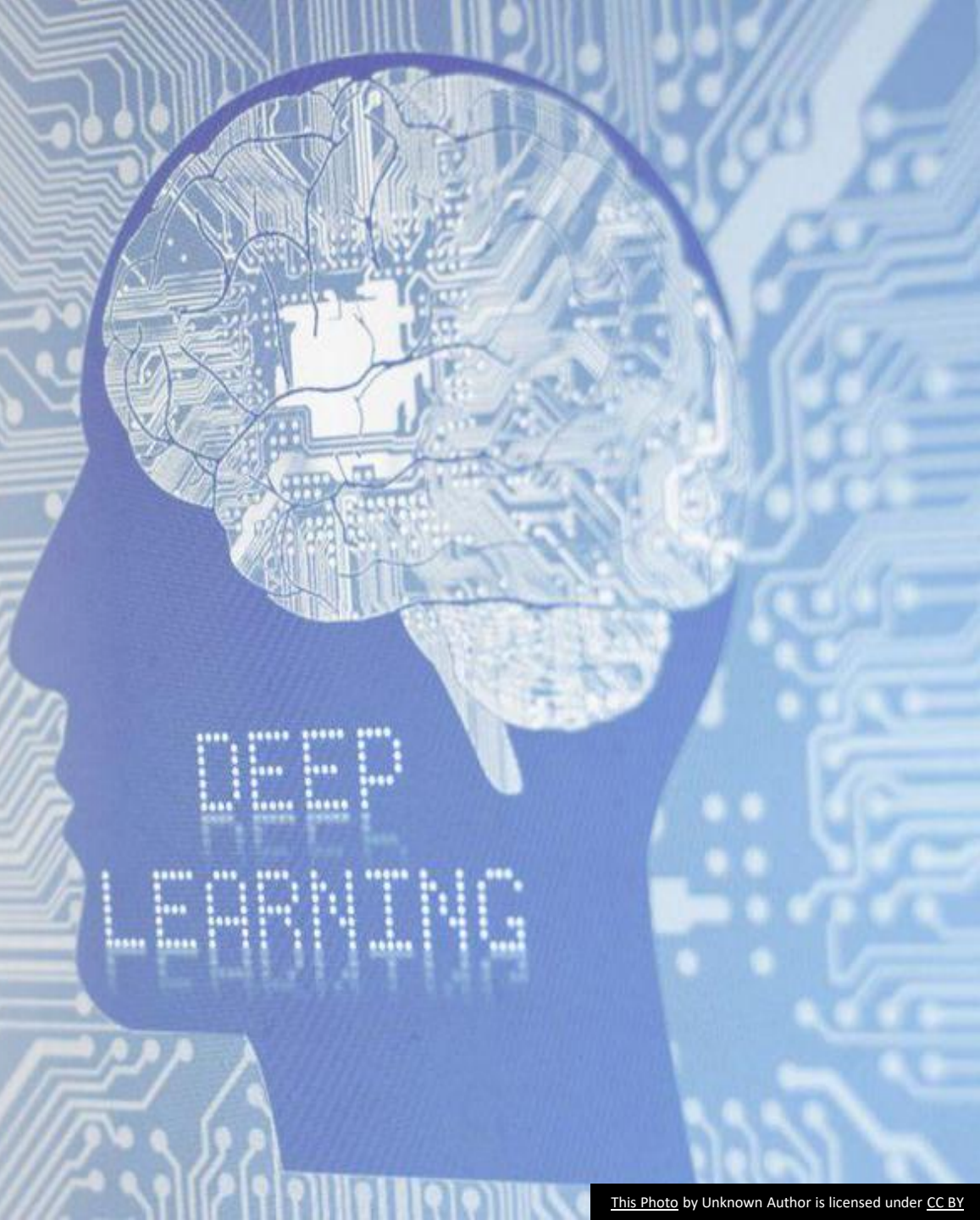

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Abstract

- This study explores a novel deep learning MetaCLIP model.
- The model leverages Vision Language Models (VLM) for skin disease image classification.
- Key features are extracted using pre-trained MetaCLIP.
- The extracted features are classified using approaches such as SVC, Ridge Classifier CV, and LDA.
- The approach is validated across 15 distinct datasets.
- Evaluation employs precision, recall, F1 score, and accuracy metrics.
- The model exhibited outstanding accuracy, attaining 90% across six datasets.

Objective

- The primary objective of this project is to evaluate 28 classifiers across 15 datasets and determine the most effective classifier, supported by the advanced MetaCLIP model, for accurate and reliable skin disease classification.

Introduction

- Skin diseases include a wide range of conditions that affect the skin, which is the largest and most visible organ of the body.
- Skin diseases can occur due to various factors like exposure to radiation, UV rays, heat rash, and other conditions, as shown in the figure.
- Initially, these diseases may appear as minor symptoms, such as red patches or rashes on the skin.
- If neglected or untreated, these early signs can escalate into severe skin diseases, including Melanoma is a dangerous form of skin cancer originating from melanocytes. Basal Cell Carcinoma (BCC) is a common form of skin cancer that grows slowly and rarely spreads. Squamous Cell Carcinoma (SCC) is a type of skin cancer that can be aggressive if not treated promptly.

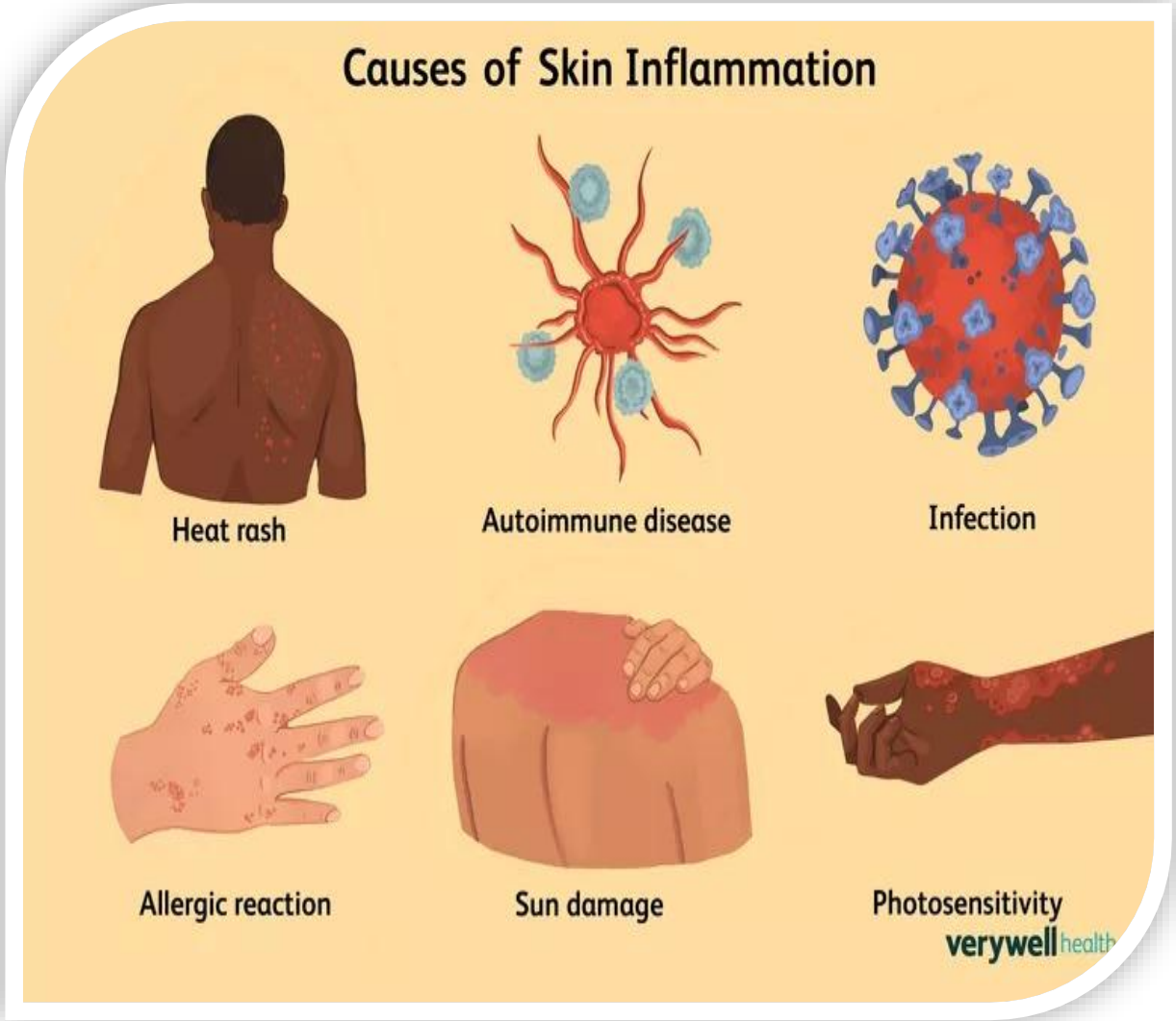


Figure 1: Cause of Skin Disease

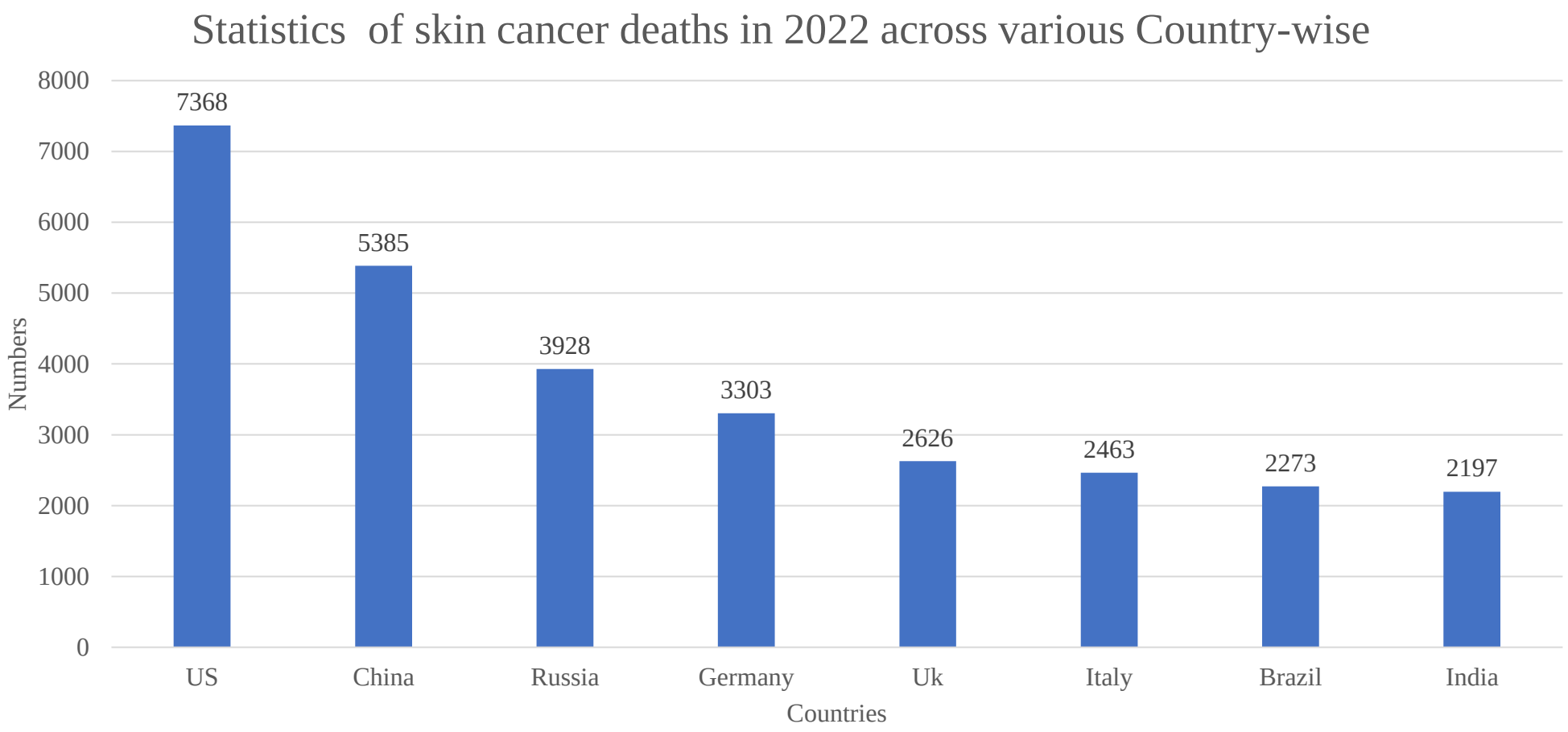


Figure 2: Number of skin cancer deaths in 2022 across various countries.

- Due to Serious skin disease remains a significant global health concern, causing over 58,000 deaths annually worldwide and nearly 2,200 deaths in India as of 2022 (World Cancer Research Fund) and requiring early diagnosis and treatment to improve patient outcomes and prevent complications.
- Skin diseases are identified by using Traditional diagnostic methods like physical exams and biopsies.
- Melanoma is typically treated with surgical removal of the tumor, and advanced stages may require therapies like immunotherapy, targeted therapy, radiation therapy, or chemotherapy.
- Basal Cell Carcinoma (BCC) and Squamous Cell Carcinoma (SCC) are treated with procedures like surgical excision or Mohs surgery, and additional treatments such as cryotherapy or radiation may be used based on the severity.

Literature Review

Table 1: Literature Review

S. No	Paper Name	Author	Year	Model	Result
1	Detection and Classification of Skin Cancer using Binary Classifier, Residual Network, and Convolutional Neural Network	Urmila Pilania, Manoj Kumar, Priyam Garg, Reet Kaur	2024	Resnet	80
2	Early Melanoma Detection and Classification Using CNN and Confusion Matrix Analysis	K. Dinesh Kumar, K. Jaya Deepthi, S. SaravanaKumar	2024	CNN	89
3	Implementation of InceptionV3 Model for Skin Disease Detection and Classification	Kunam Subba Reddy, B.Lalithadevi	2025	InceptionV3	90
4	Enhanced skin cancer diagnosis using optimized CNN architecture and checkpoints for automated dermatological lesion Classification	M. Mohamed Musthafa, Mahesh, Vinoth Kumar, Suresh Guluwadi	2024	CNN	97

Literature Review (Cont..)

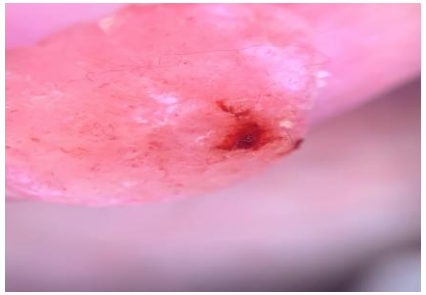
S. No	Paper Name	Author	Year	Model	Result
5	Learning Transferable Visual Models From Natural Language Supervision	Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G.	2021	CLIP	-
6	Demystifying CLIP Data	Hu Xu, Saining Xie, Xiaoqing Ellen Tan, Po-Yao Huang, Russell Howes, Vasu Sharma, Shang-Wen Li	2023	MetaCLIP	-

Problem Statement

- Traditional diagnostic methods like physical exams and biopsies are often time-consuming and unreliable for early detection.
- CNNs help speed up diagnosis but have limitations. They mainly concentrate on local features, which may lead to overlooking broader patterns essential for accurate diagnosis. Additionally, CNNs rely on pre-processed images for classification since variations in lighting, angles, and image quality can greatly affect their performance.
- CLIP analyzes the broader patterns and links images with text, but it struggles with fine details in medical images, like subtle skin lesion differences.
- ViT-H-14metaclip_v1_2_altogether from the MetaCLIP family addresses these limitations by focusing on both the broader patterns and fine details, while not requiring pre-processed images, and maintaining image-text relationships.

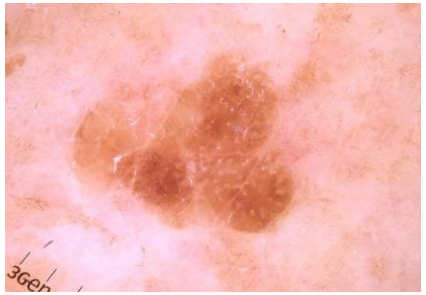
Datasets Description

Vascular lesion

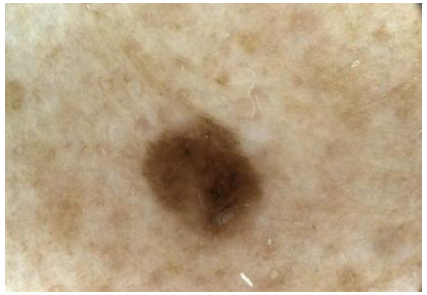


Squamous cell carcinoma

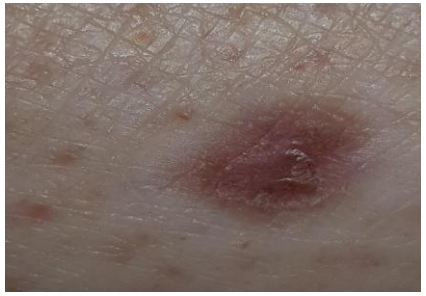
Seborrheic keratosis



Pigmented benign keratosis



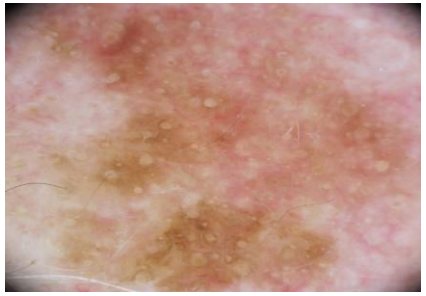
Nevus



Dermatofibroma

Basal cell carcinoma

Melanoma



Actinic keratosis

Bullous disease

Figure 3: Sample Images from the 15 datasets

Datasets Description (Cont..)

Table 2: Dataset table for the 15 datasets

S.NO	Datasets Names	No. of Classes	Total No .of Images
1	Skin cancer HAM 10000 Hair removal	7	10015
2	Multiple skin diseases detection	9	4109
3	Skin cancer ISIC	9	2375
4	Fluo-sc	6	3146
5	dermnet	23	19559
6	Skin Disease Dataset	9	5400
7	DermPK	6	157
8	Natural Data-augmentation for skin Lesions	3	8100
9	skin disease identification	5	7357
10	skin_disease	6	1462
11	Melanoma skin cancer dataset of 10000 images	2	10605
12	5-Common-human -skin-disease	6	3295
13	skin-disease-Cancer	2	900
14	Monkeypox Skin images dataset(MSID)	4	770
15	Skin.Cancer.data	2	3297

Datasets Description (Cont..)

Table 3: Sample Class Names Across 5 Datasets

S. No	Name of the Dataset	No. of classes	Name of classes	No .of Images
1	Skin cancer HAM10000	7	1. Actinic keratosis and intraepithelial carcinoma 2. Basal cell carcinoma 3. Benign keratosis-like lesions 4. Dermatofibroma 5. Melanoma 6. Melanocytic nevi 7. Vascular lesions	10015

Datasets Description (Cont..)

2	Fluo-sc	6	1. Basal Cell Carcinoma 2. Squamous Cell Carcinoma 3. Melanoma 4. Actinic Keratosis 5. Seborrheic Keratosis 6. Nevus	3146
3	DermPK	6	1. Melanoma 2. Melanocytic Nevus 3. Basal Cell Carcinoma 4. Squamous Cell Carcinoma 5. Benign Keratosis 6. Unknown	157

Datasets Description (Cont..)

4	Natural Data-augmentation for skin Lesions (ISIC-2017 Challenge dataset)	3	1. Melanoma 2. Nevus 3. Skin Lesion,	8100
5	5-Common-human -skin-disease	6	1. Acne An Rosacae 2. ChickenPox 3. Eczema 4. Normal 5. Ringworm 6. Seborrheic Keratoses	3295

MetaCLIP Architecture

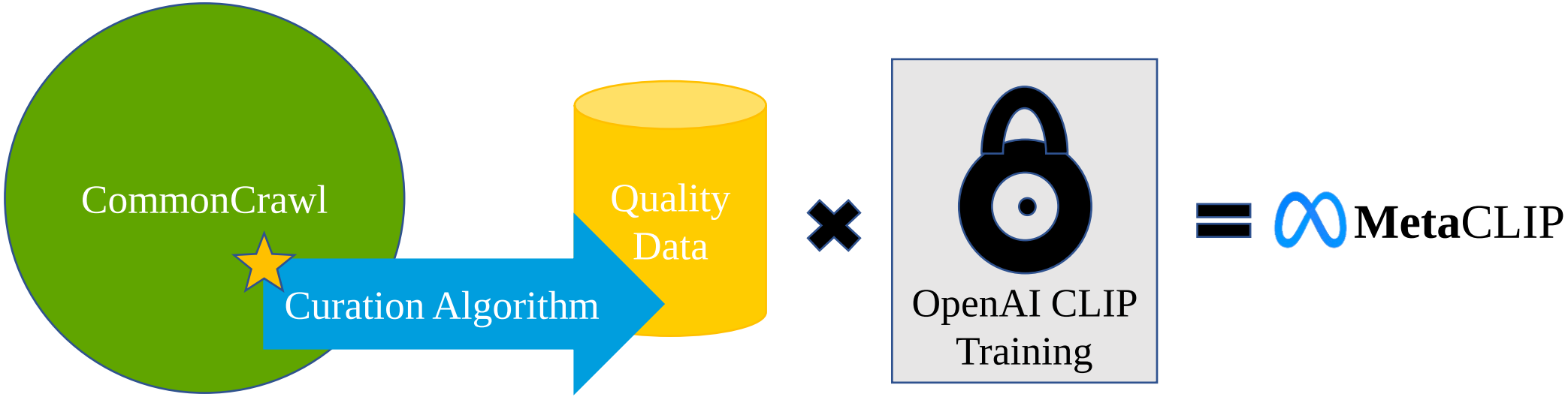


Figure 4: MetaCLLIP Architecture

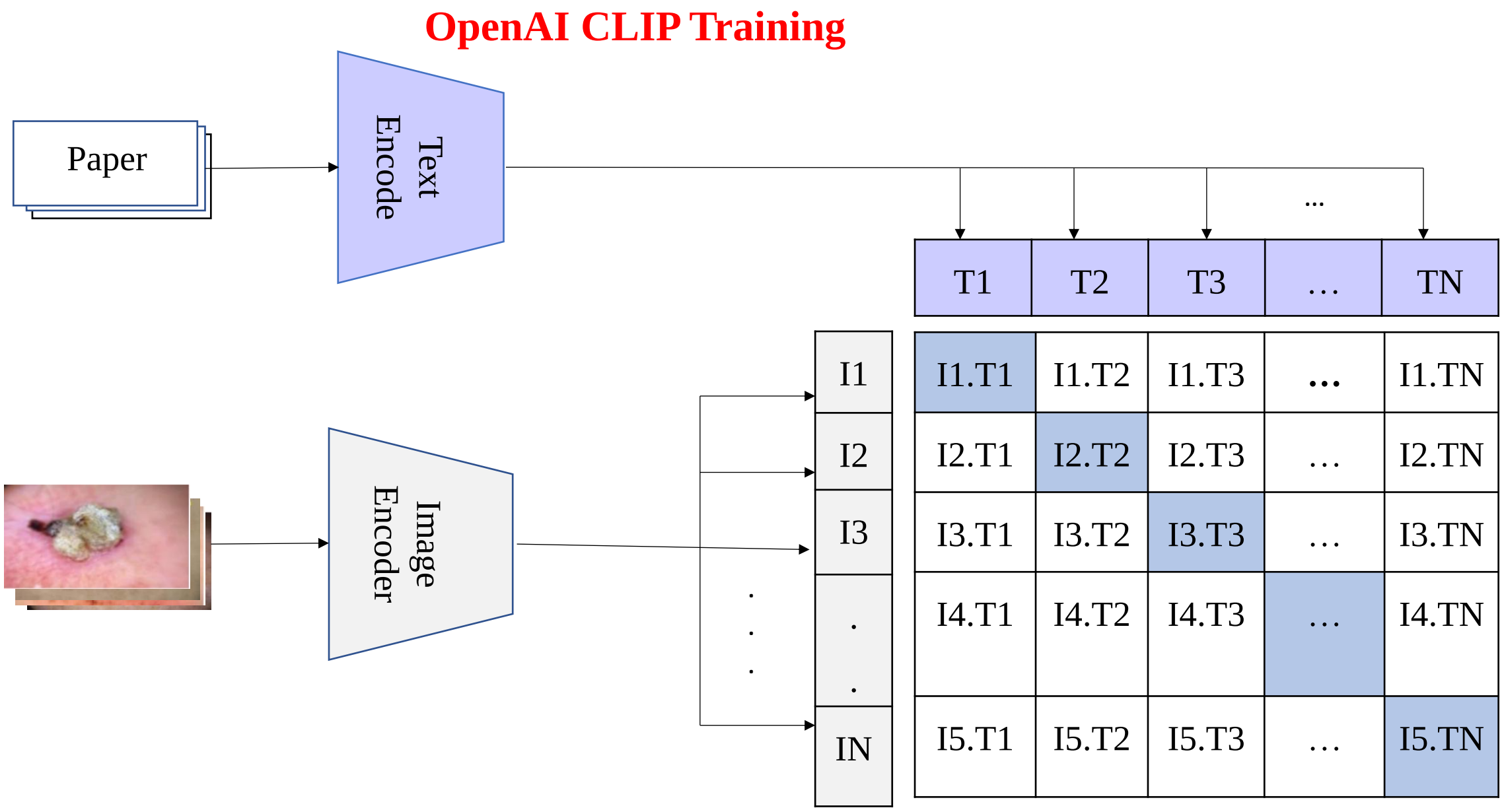


Figure 5: OpenAI Training

Proposed Algorithm

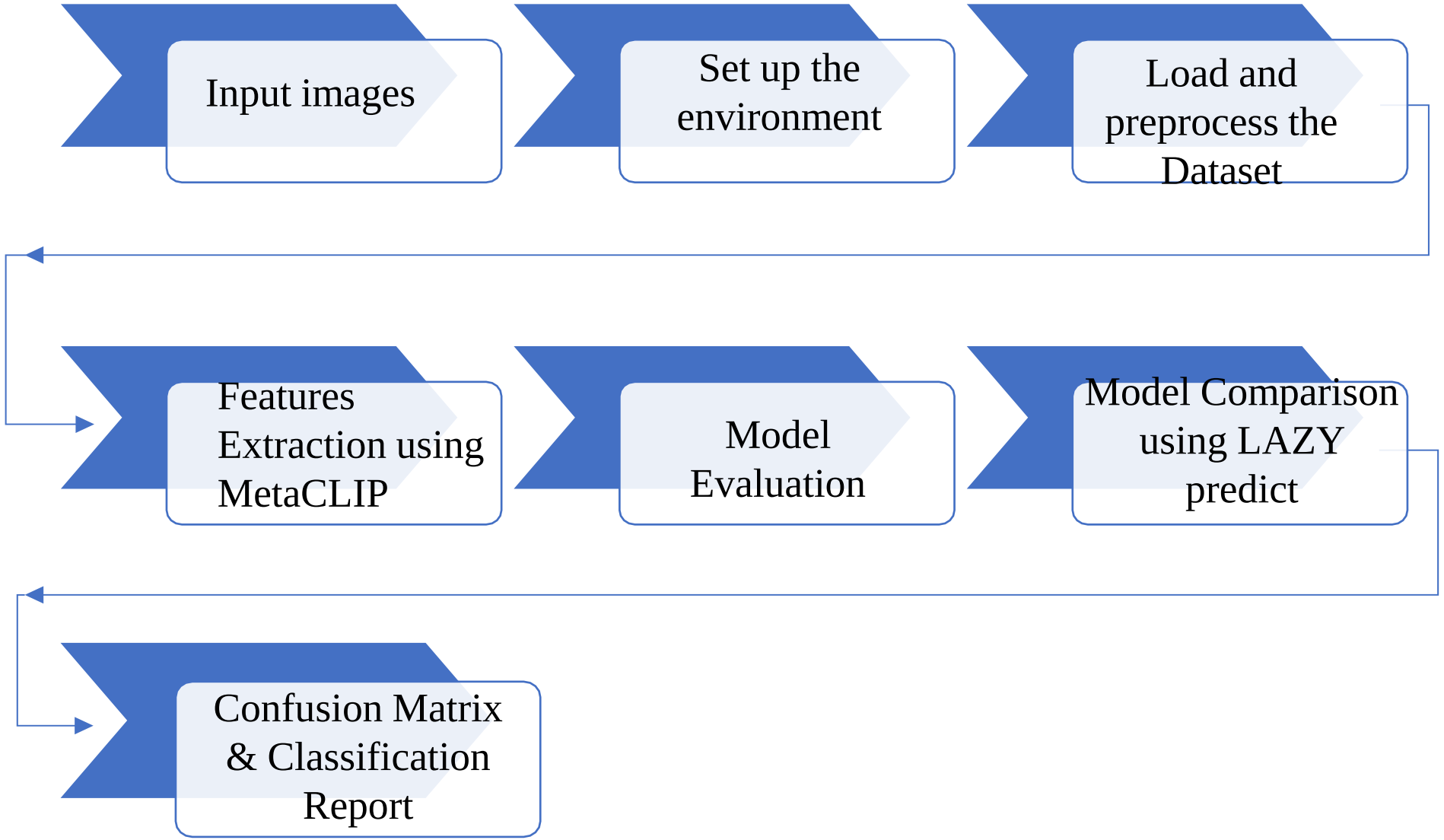


Figure 6: Proposed Algorithm

Result and Simulations

Table 2: Classification report for 5-Common –Human-Skin-Disease dataset

S. No	Classifiers	Accuracy	Balanced Accuracy	Time Taken
1	SVC	0.9939	0.9947	1.38200
2	XGB Classifier	0.9939	0.9947	20.5982
3	LGBM Classifier	0.9939	0.9947	60.7446
4	SGD Classifier	0.9939	0.9934	0.54469
5	Extra Trees Classifier	0.9909	0.9927	1.02946
6	LinearSVC	0.9909	0.9914	30.6573
7	Logistic Regression	0.9909	0.9914	0.38892
8	Perceptron	0.9909	0.9914	0.49044
9	PassiveAggressiveClassifier	0.9909	0.9914	0.85327
10	Calibrated Classifier CV	0.9909	0.9914	158.741
11	Bagging Classifier	0.9878	0.9907	17.1464
12	Kneighbors Classifier	0.9878	0.9907	0.18473
13	Ridge Classifier	0.9848	0.9874	0.36308

Results and Simulations(Cont..)

S.No	Classifiers	Accuracy	Balanced accuracy	Time Taken
14	RidgeClassifier	0.9848	0.9874	6.59685
15	RandomForestClassifier	0.9848	0.9874	2.65991
16	RidgeClassifierCV	0.9848	0.9874	1.36034
17	LinearDiscriminantAnalysis	0.9848	0.9874	0.16201
18	NearestCentroid	0.9757	0.9826	5.48639
19	NuSVC	0.9757	0.9765	0.25547
20	BernoulliNB	0.9696	0.9748	0.14000
21	GaussianNB	0.9696	0.9748	3.44857
22	DecisionTreeClassifier	0.9515	0.9516	0.12207
23	ExtraTreeClassifier	0.9272	0.9277	34.7470
24	AdaBoostClassifier	0.8666	0.8611	1.27465
25	QuadraticDiscriminantAnalysis	0.3909	0.4038	0.53829
26	LabelPropagation	0.2242	0.2346	0.67936
27	LabelSpreading	0.2242	0.2346	6.59685

Results and Simulations(Cont..)

In the results part, we are using evaluation metrics, or performance metrics, such as Precision, Recall, F1-score, Specificity, Geometric mean, Balanced Accuracy, and Accuracy. Formulas are:

- Precision = $TP \div (TP + FP)$
- Recall =$TP \div (TP + FN)$
- F1 score = $2x(Precision\ x\ Recall\ \div Precision + Recall)$
- Specificity= $TN \div (TN + FP)$
- G-mean=$\sqrt{Recall}\ x\ Specificity$
- Balanced Accuracy= $Recall + Specificity \div 2$
- Accuracy =$TP + TN \div (TP + TN + FP + FN)$

Results and Simulations(Cont..)

Evaluation Metrics for the 15 datasets of 80-20% Cross-Validation

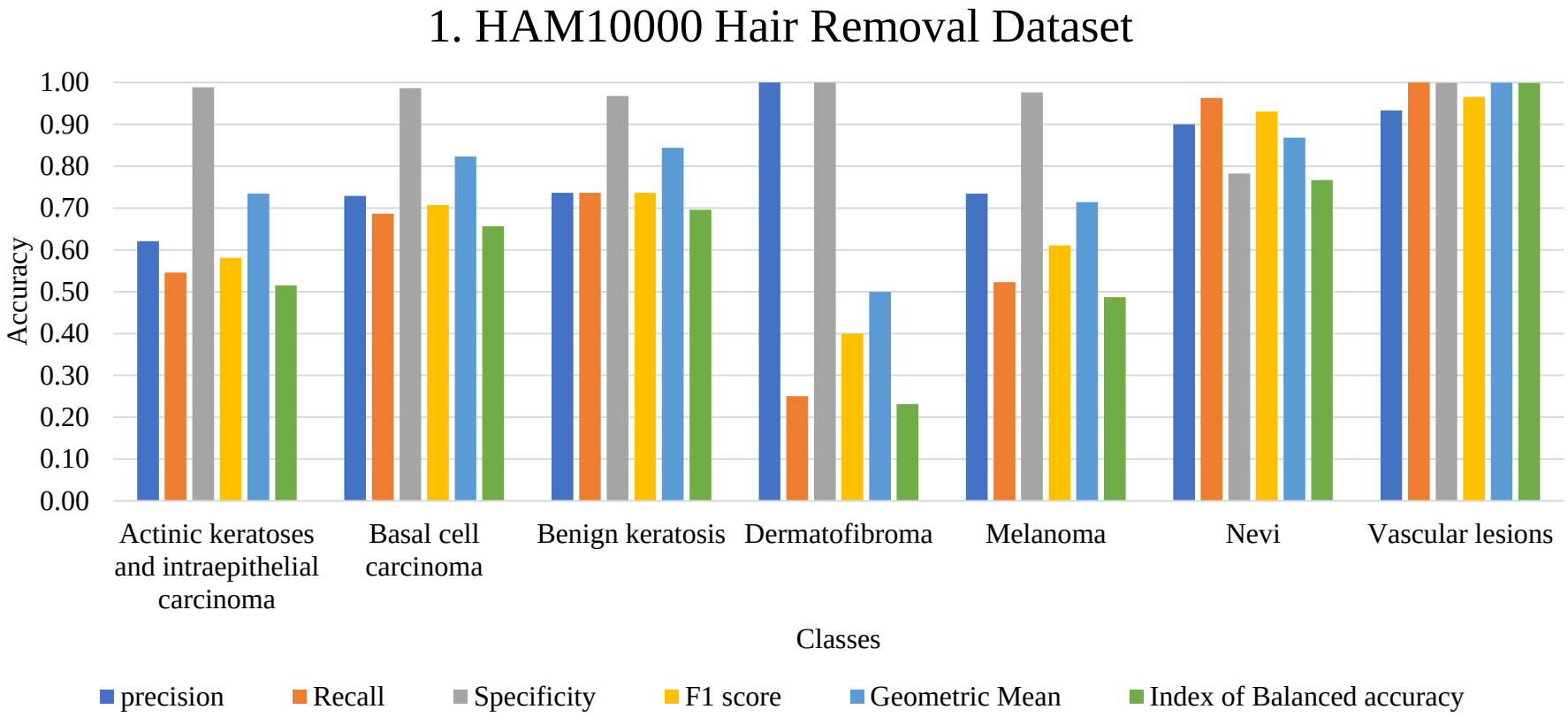


Figure 7: Evaluation Metrics for the Ham10000

Results and Simulations(Cont..)

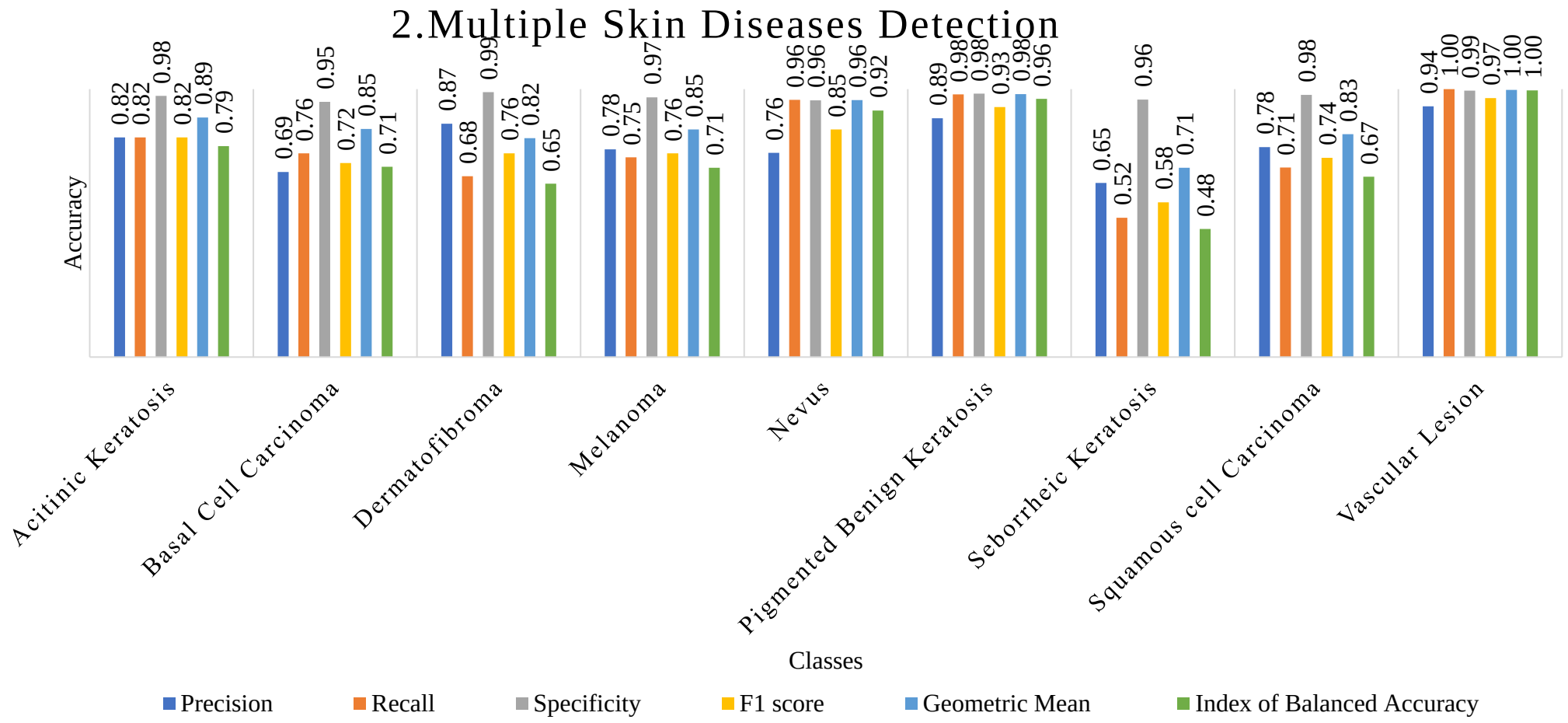


Figure 8: Evaluation Metrics for the Multiple Skin Disease Detection

Results and Simulations(Cont..)

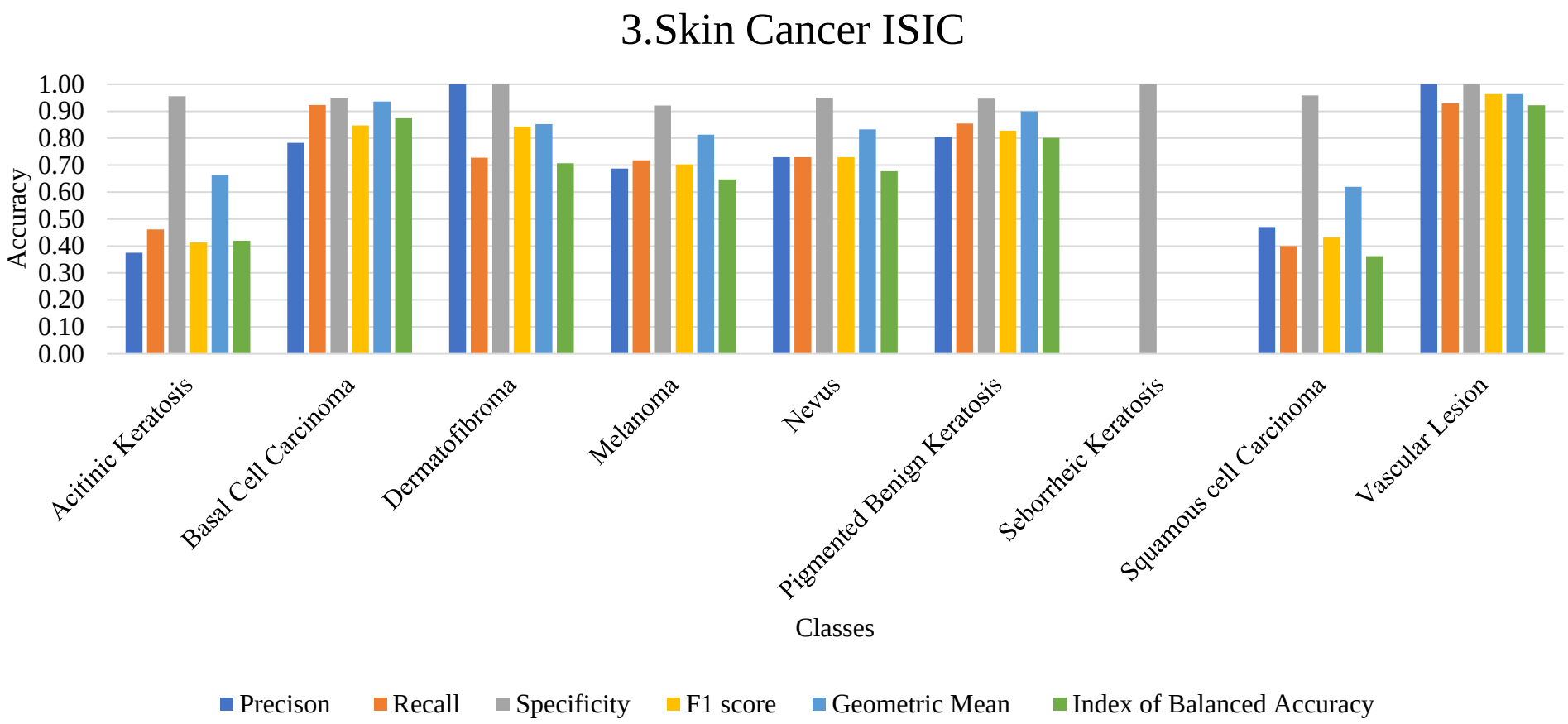


Figure 9: Evaluation Metrics for Skin Cancer ISIC

Results and Simulations(Cont..)

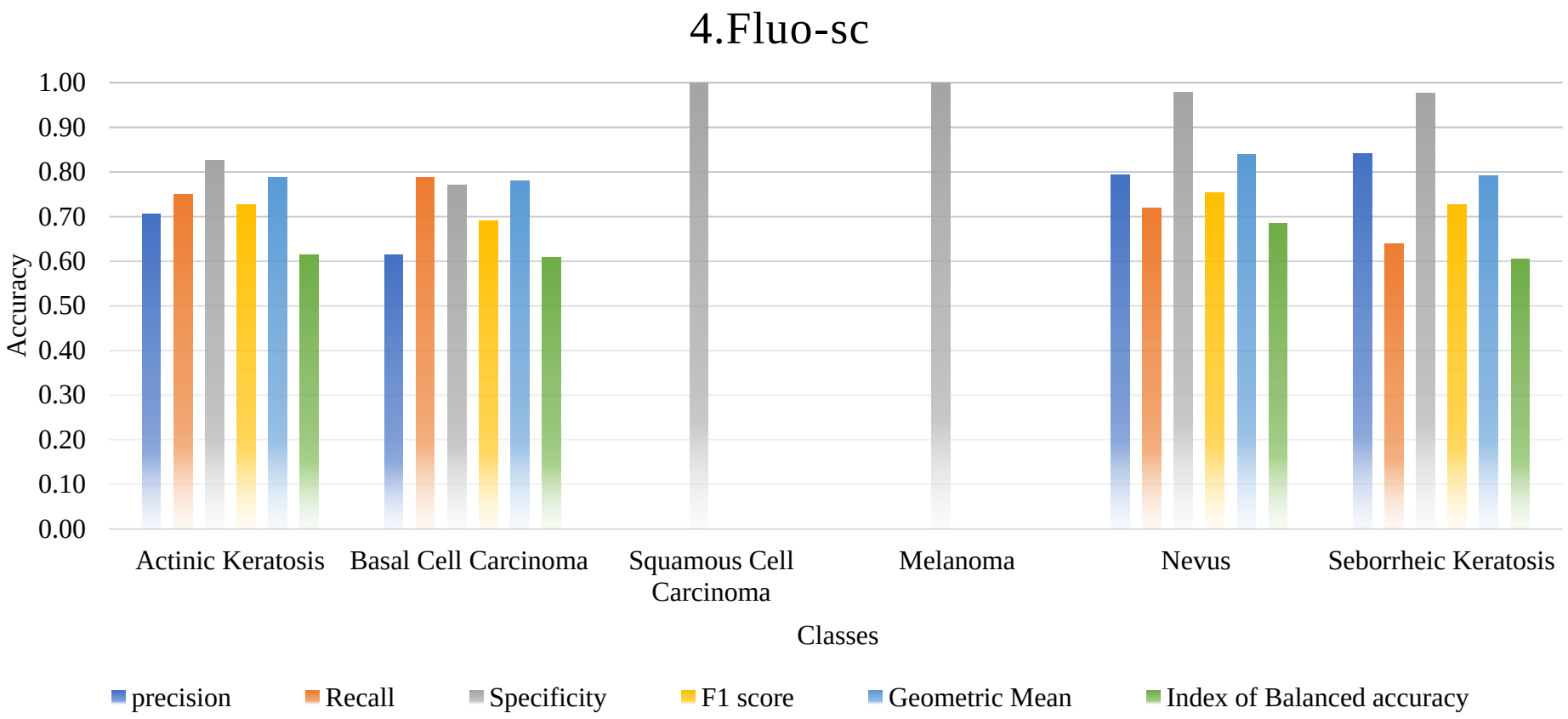


Figure 10: Evaluation Metrics for the Fluo-sc

Results and Simulations(Cont..)

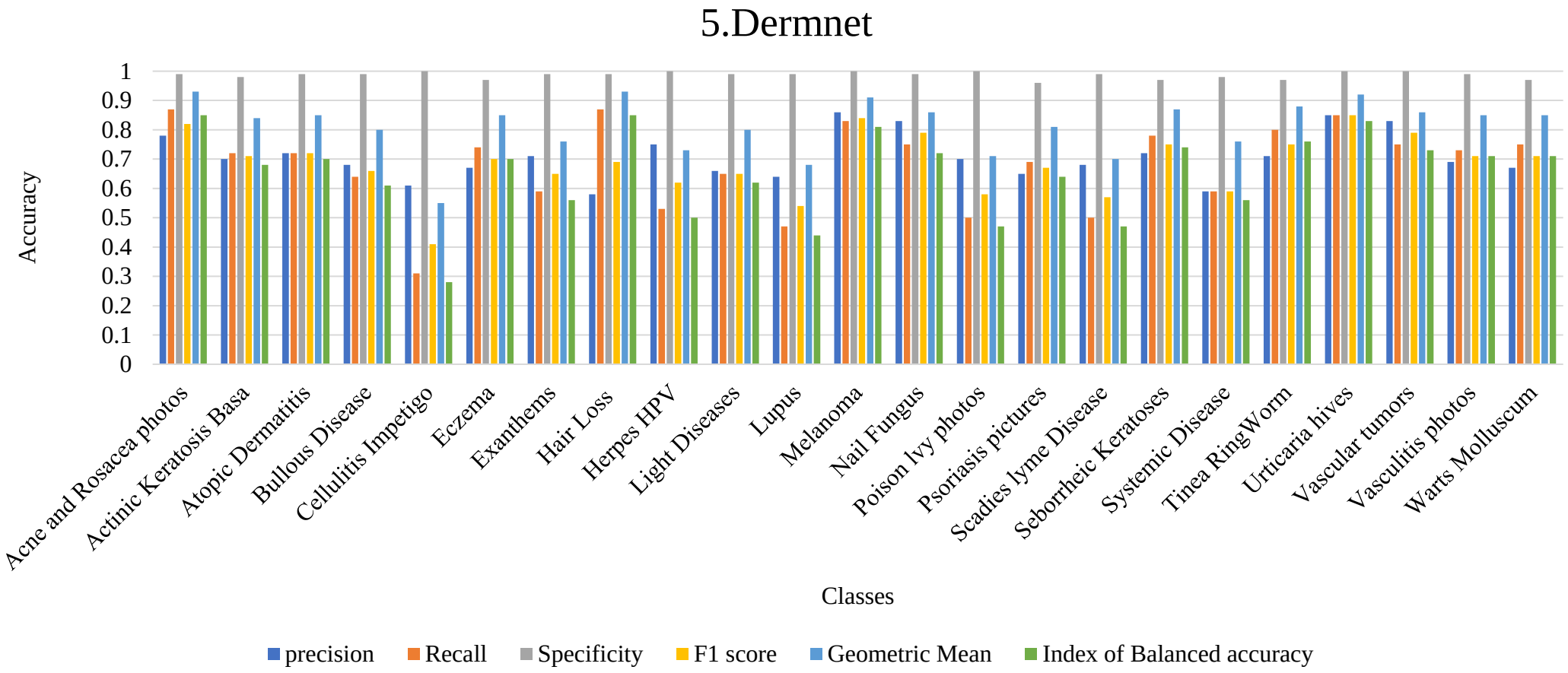


Figure 11: Evaluation Metrics for the Dermnet

Results and Simulations(Cont..)

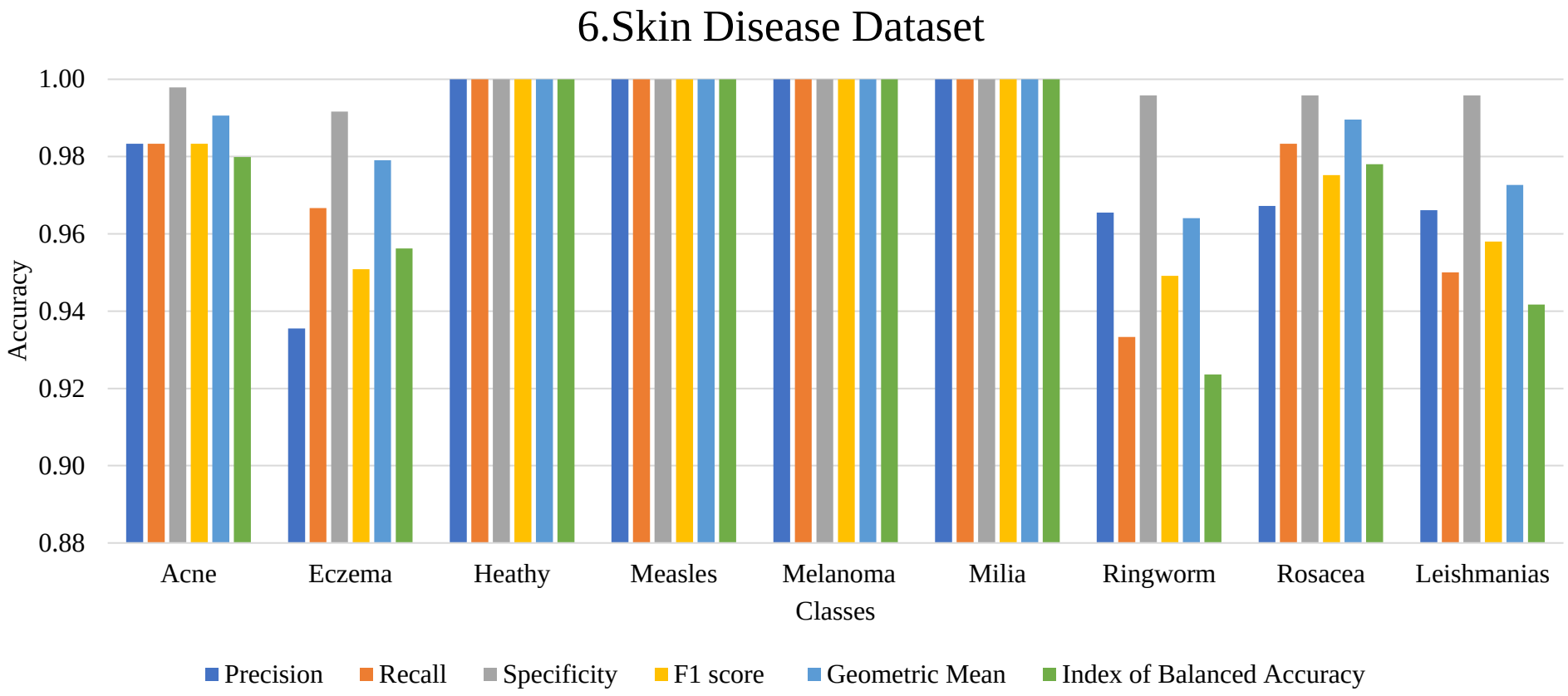
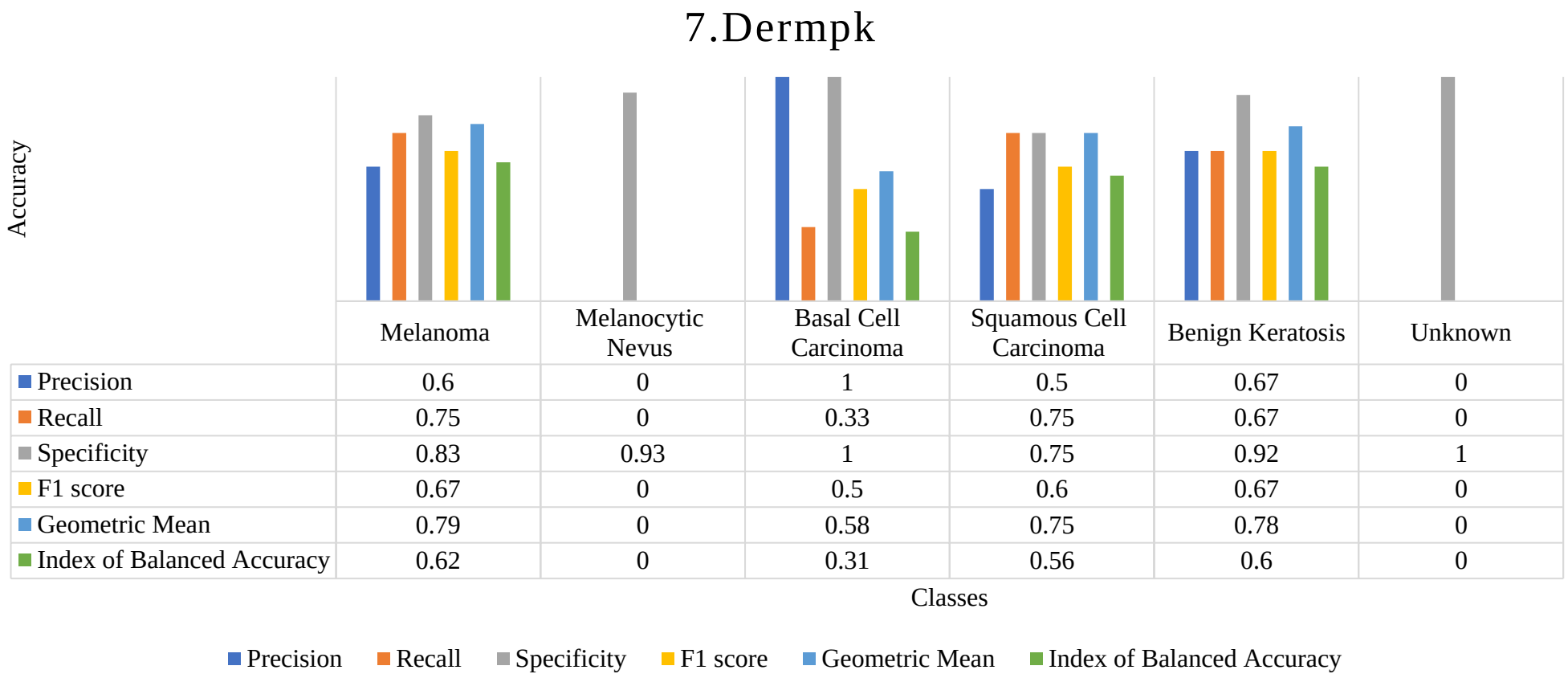


Figure 12: Evaluation Metrics for the Skin Disease Dataset

Results and Simulations(Cont..)



	Melanoma	Melanocytic Nevus	Basal Cell Carcinoma	Squamous Cell Carcinoma	Benign Keratosis	Unknown
Precision	0.6	0	1	0.5	0.67	0
Recall	0.75	0	0.33	0.75	0.67	0
Specificity	0.83	0.93	1	0.75	0.92	1
F1 score	0.67	0	0.5	0.6	0.67	0
Geometric Mean	0.79	0	0.58	0.75	0.78	0
Index of Balanced Accuracy	0.62	0	0.31	0.56	0.6	0

Figure 13: Evaluation Metrics for the DermPK

Results and Simulations(Cont..)

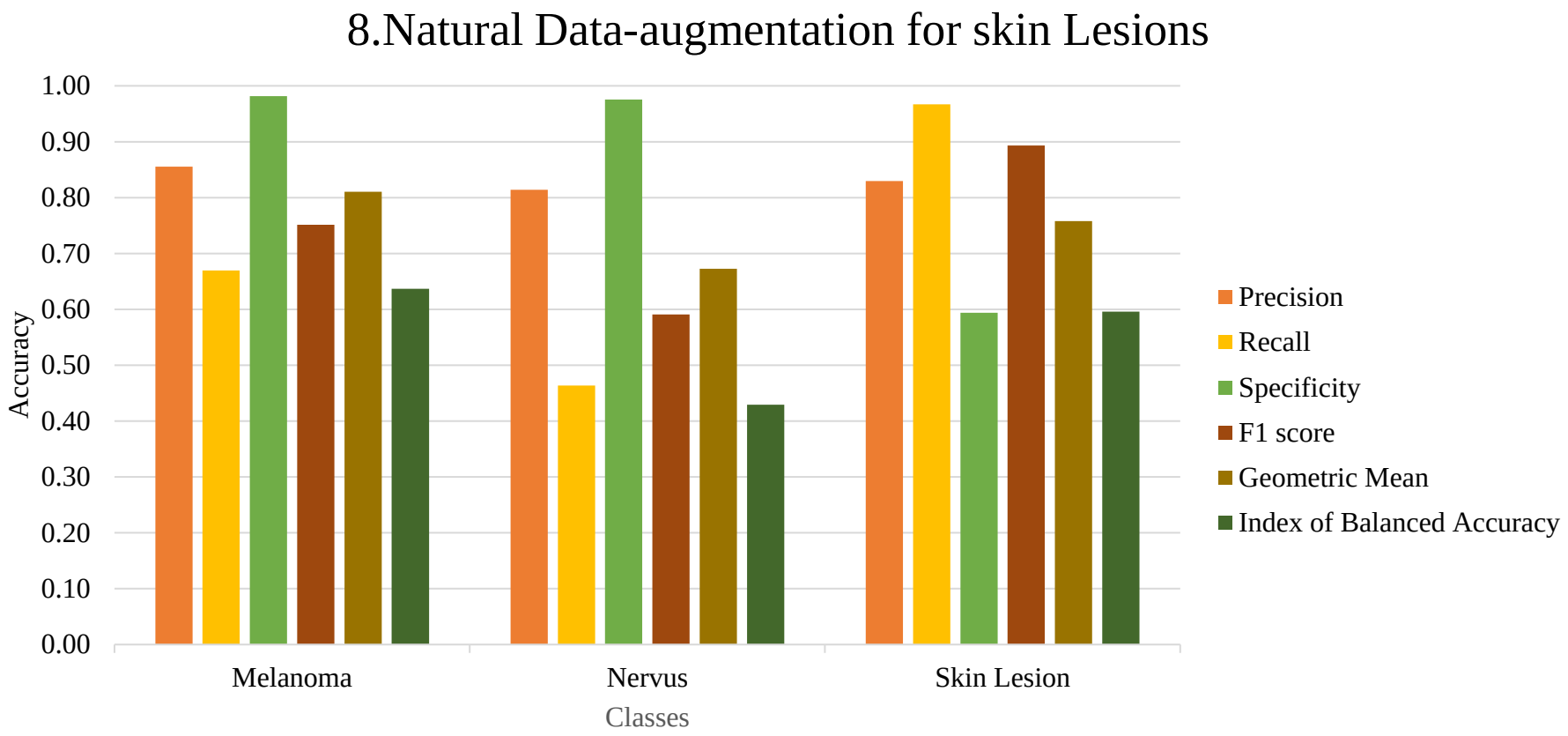


Figure 14: Evaluation Metrics for the Natural Data-augmentation for Skin Lesions

Results and Simulations(Cont..)

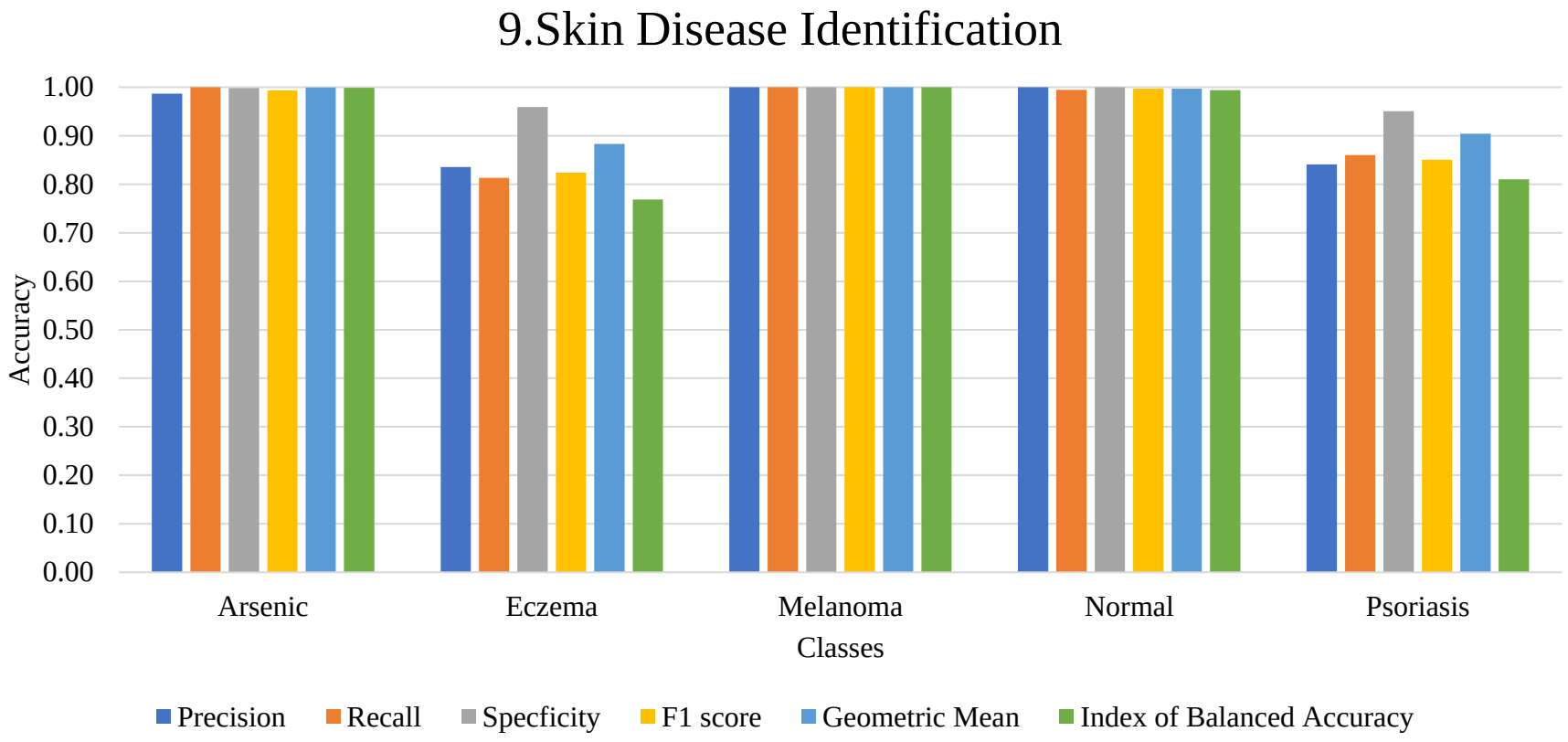


Figure 15: Evaluation Metrics for the Skin Disease Identification

Results and Simulations(Cont..)

10.Skin-Disease

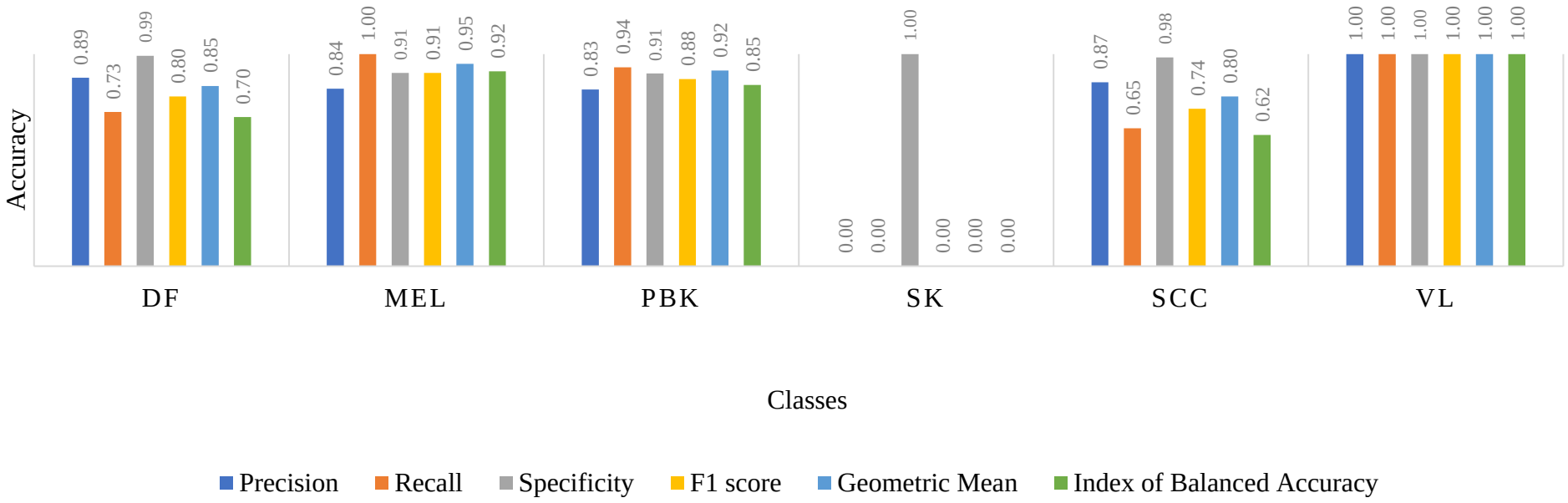


Figure 16: Evaluation metrics for the Skin_Disease

Results and Simulations(Cont..)

11.Melanoma Skin Cancer Dataset of 10000 Image

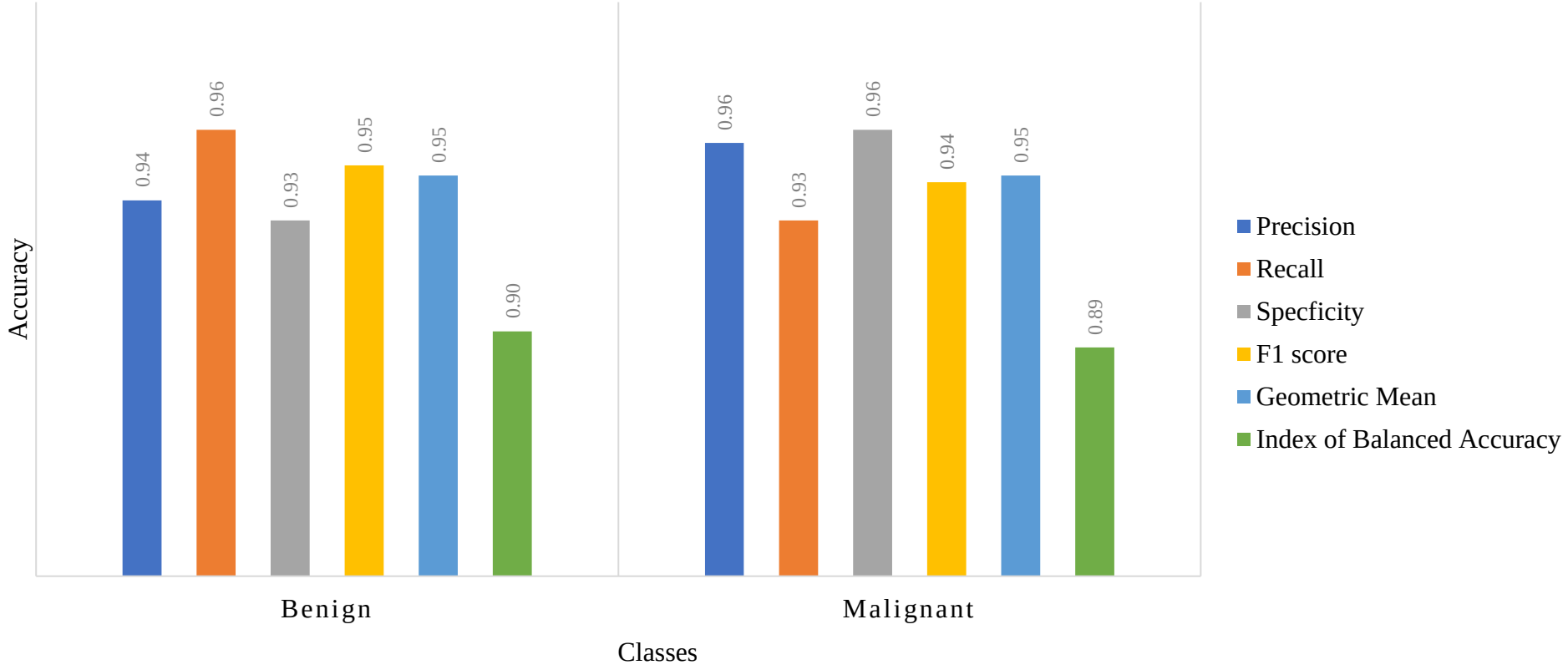


Figure 17: Evaluation metrics for the Melanoma Skin cancer of 10000 images

Results and Simulations(Cont..)

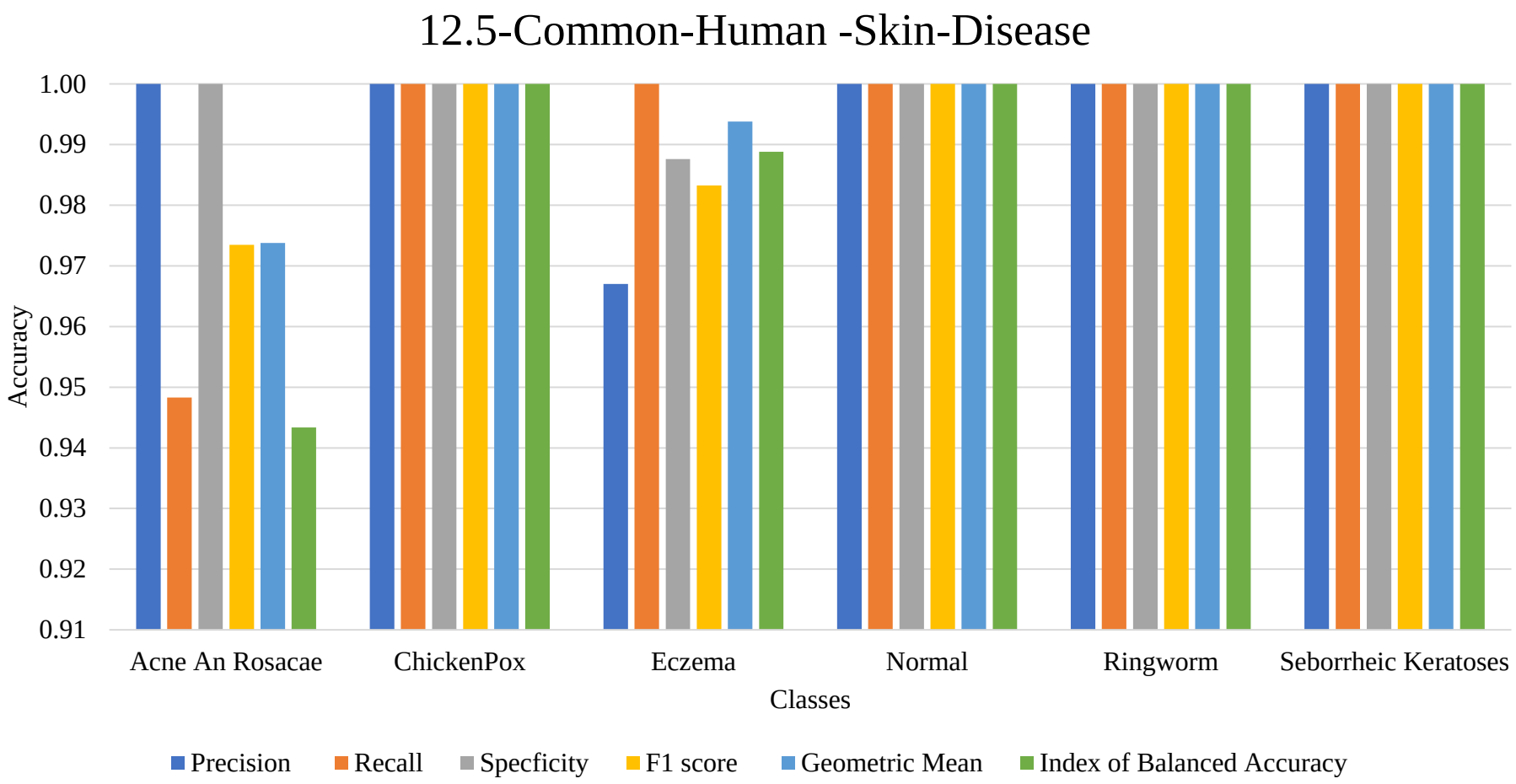


Figure 18: Evaluation Metrics for the 5-Common-Human-Skin-Diseases

Results and Simulations(Cont..)

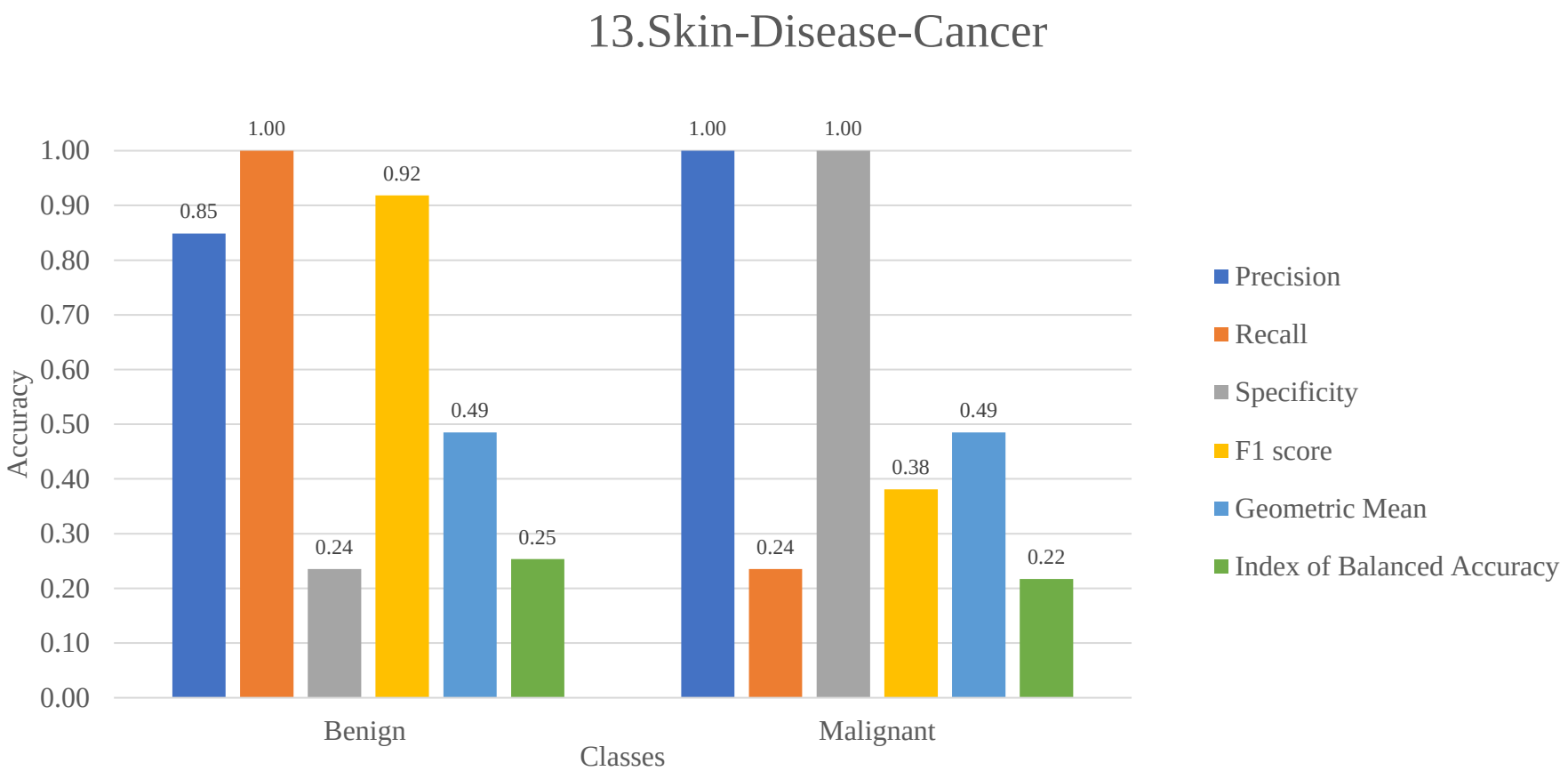


Figure 19: Evaluation Metrics for the Skin-Disease-Cancer

Results and Simulations(Cont..)

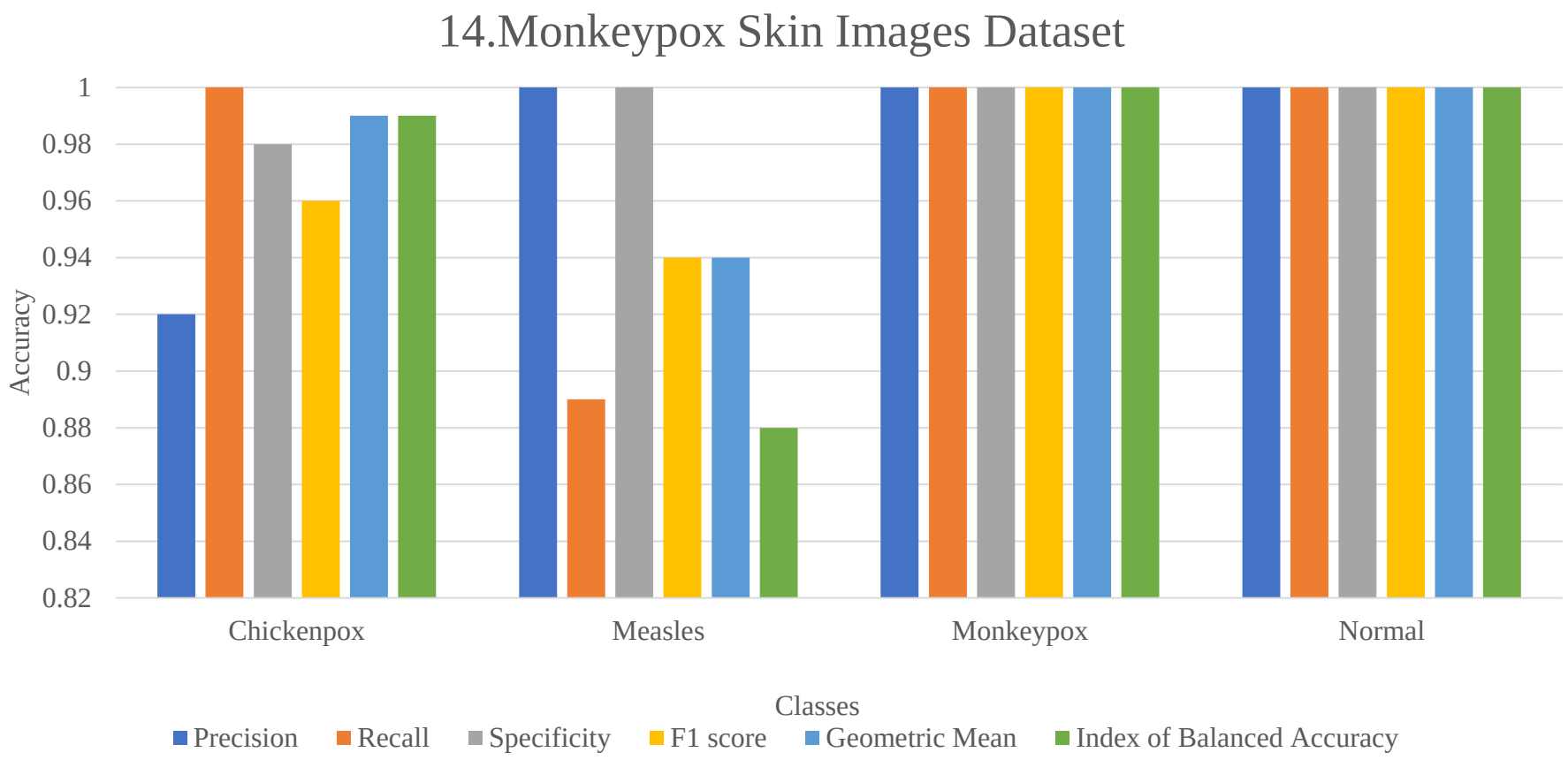


Figure 20: Evaluation Metrics for Monkeypox skin images

Results and Simulations(Cont..)

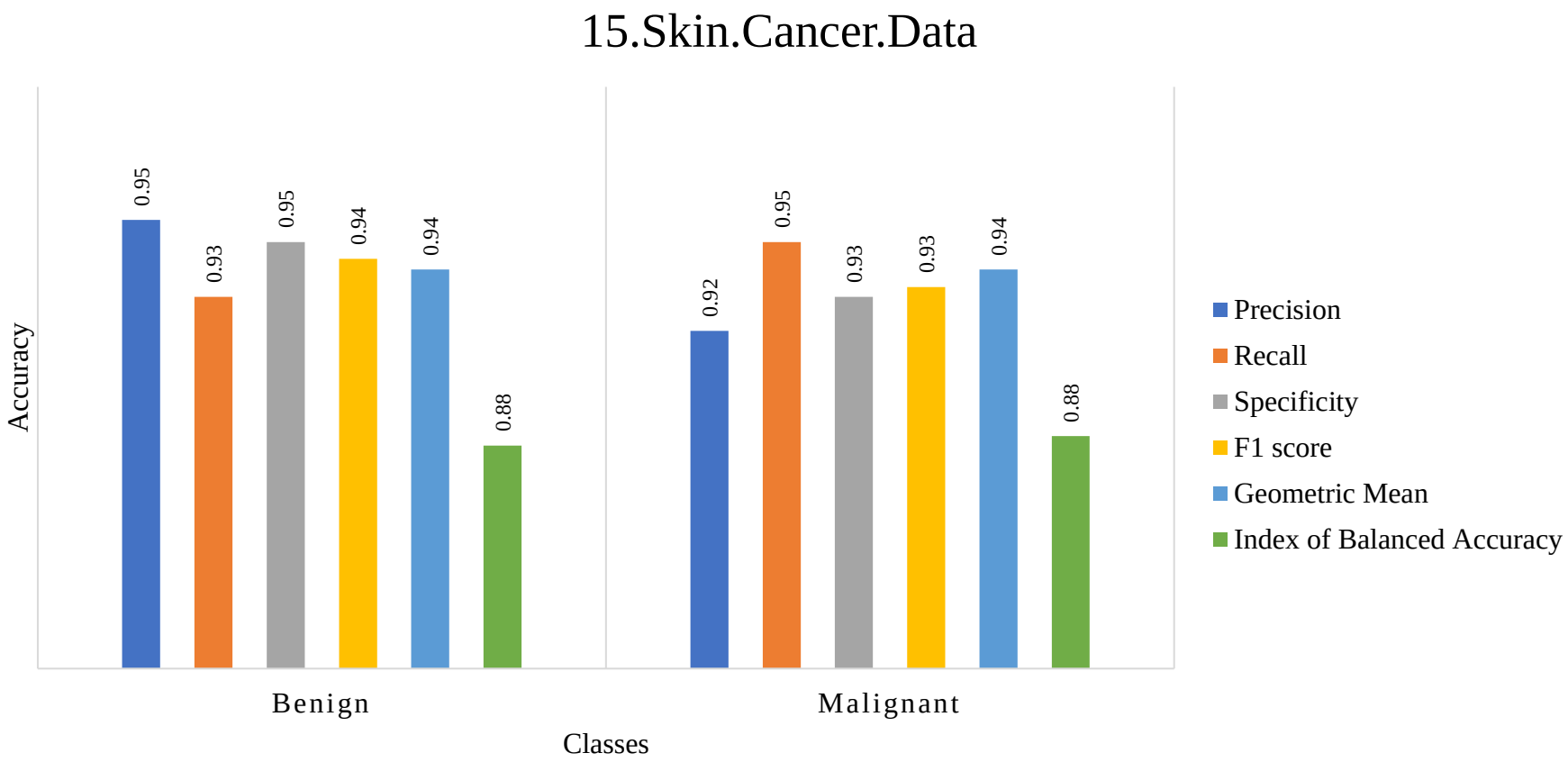


Figure 21: Evaluation Metrics for the Skin.Cancer.Data

Results and Simulations(Cont..)

Accuracy Comparison

Table 4 : Accuracy comparison of 15 datasets across (80-20) % and (90-10) % cross -validation

S.NO	Dataset Names	Classifiers	Accuracy	
			Cross-validation (80-20)%	Cross-validation (90-10)%
1	Skin cancer HAM 10000 Hair removal	Linear Discrimination Analysis	73	74
2	Multiple skin diseases detection	SVC	76	77
3	Skin cancer ISIC	SVC	63	63
4	Fluo-sc	SVC	51	51
5	dermnet	SVC	57	70
6	Skin Disease Dataset	SVC	97	97
7	DermPK	Ridge Classifier CV	78	79
8	Natural Data-augmentation for skin Lesions	SVC	71	71
9	skin disease identification	LGBM classifier	95	95.8
10	skin_disease	SVC	69	70

Results and Simulations(Cont..)

S.NO	Dataset Names	Classifiers	Accuracy	
			Cross-validation (80-20)%	Cross-validation (90-10)%
11	Melanoma skin cancer dataset of 10000 images	SVC	93	94
12	5-Common-human -skin-disease	SVC	99	99
13	skin-disease-Cancer		70	70
14	Monkeypox Skin images dataset(MSID)	SVC	95	95
15	Skin.Cancer.data	SVC	90	90

Conclusion

- ✓ The MetaCLIP deep learning model, specifically ViT-H-14metaclip_v1_2_altogether, was successfully used for feature extraction in skin disease classification.
- ✓ A total of 15 datasets were analyzed, with 6 datasets achieving over 90% accuracy in classification.
- ✓ The SVC classifier performed exceptionally well, achieving 99% accuracy for the 5-Common-Human-Skin-Diseases dataset.
- ✓ MetaCLIP demonstrated high precision, recall, and balanced accuracy, outperforming traditional AI models in dermatological diagnosis.
- ✓ The study concluded that MetaCLIP, when combined with the SVC classifier, achieved the best performance on skin disease images.
- ✓ Beyond skin disease classification, MetaCLIP can be applied to other domains, such as traffic management, retail object detection, and anomaly detection in manufacturing systems.

References

1. U. Pilania, M. Kumar, P. Garg, and R. Kaur, "Detection and classification of skin cancer using binary classifier, residual network, and convolutional neural network," Proceedings of the 2nd International Conference on Sustainable Computing and Smart Systems (ICSCSS 2024), pp. 1-6, 2024, DOI: 10.1109/ICSCSS60660.2024.10625032.

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THANK YOU